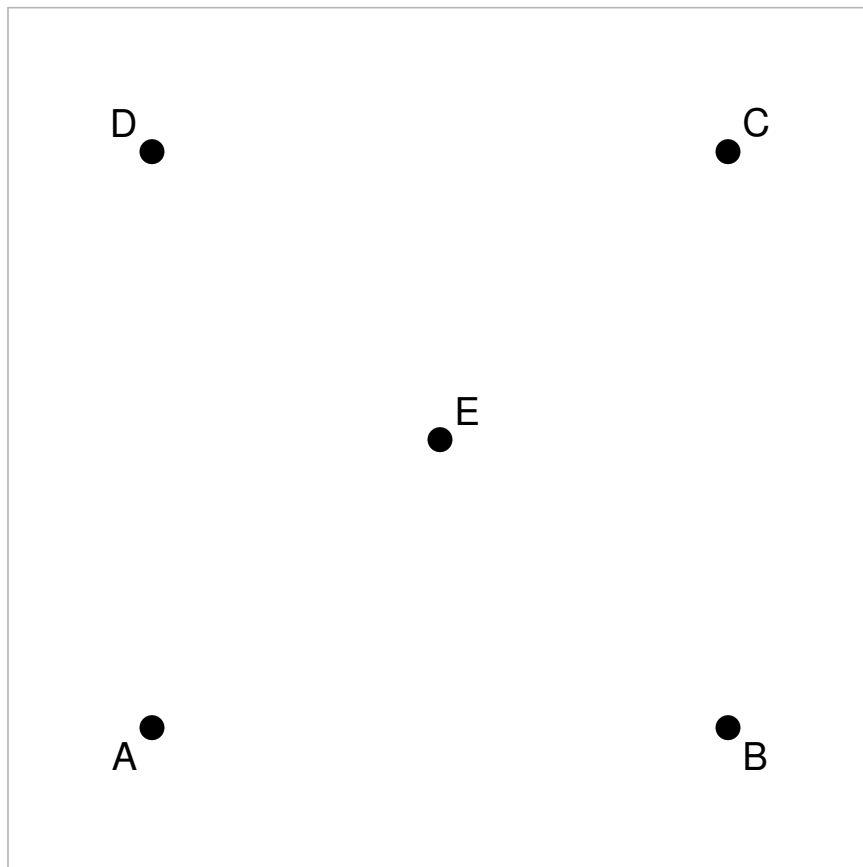
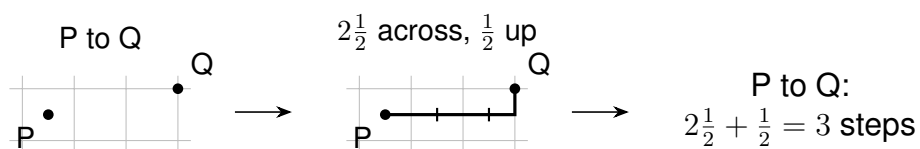


Use string to compare straight-line distances. A tie gets all tied letters. The map continues beyond the frame.

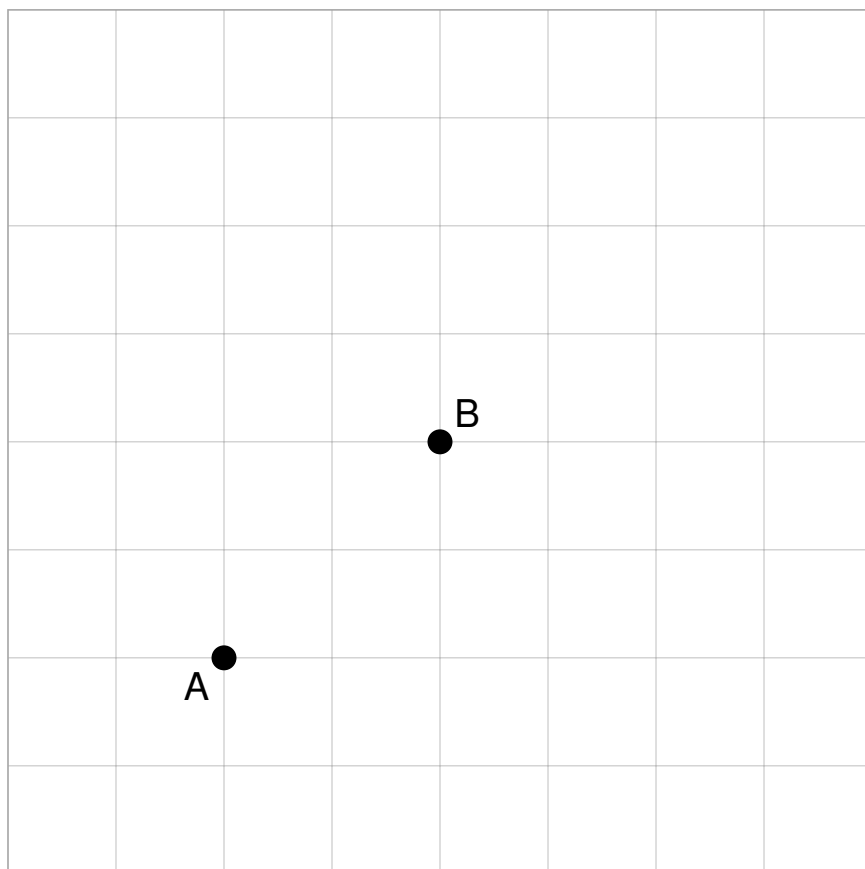
Problem 1: Mark and label places with the letter of the dot farthest away. Can each dot be farthest from somewhere?



Travel horizontally or vertically from any point, including inside squares. One grid edge is one step; a fraction of an edge is that fraction of a step. Distance is the length of a shortest trip. A tie gets both letters. The grid continues beyond the frame.

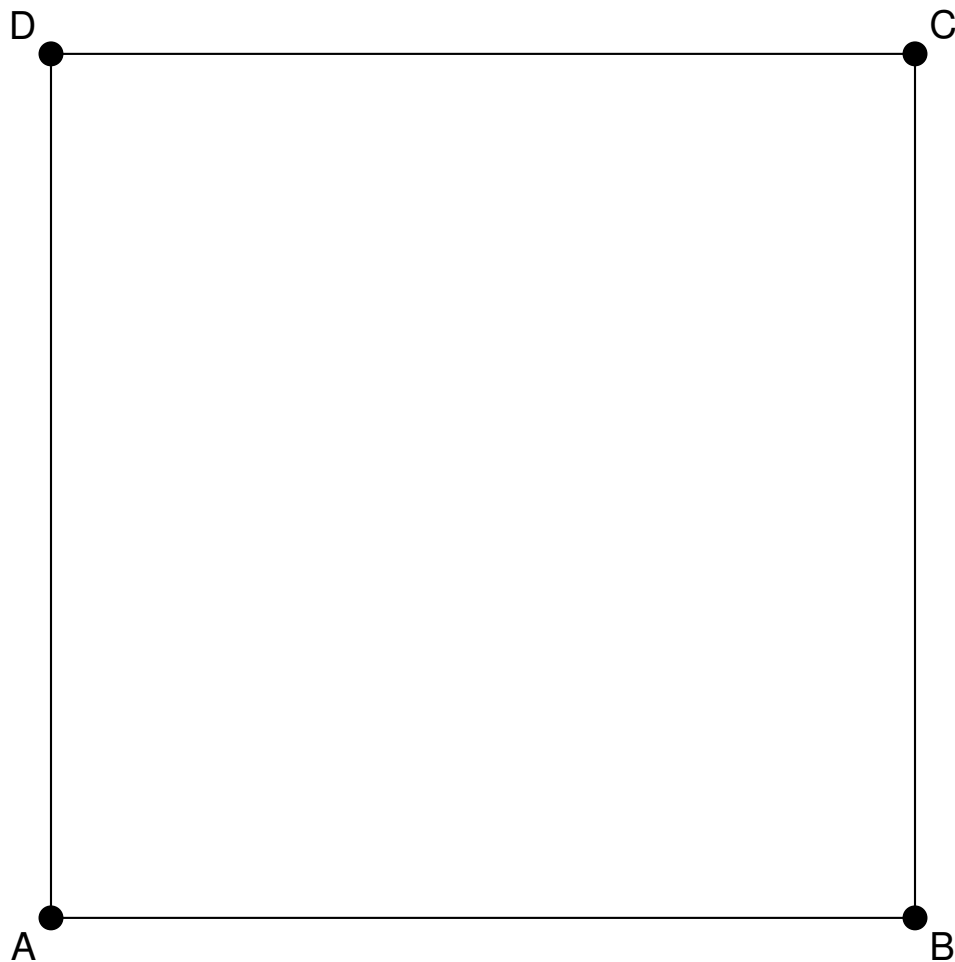


Problem 2: Label grid crossings A, B, or AB for the nearer dot. Do ties fill whole small squares? Can two tied places have a midpoint that is not tied?



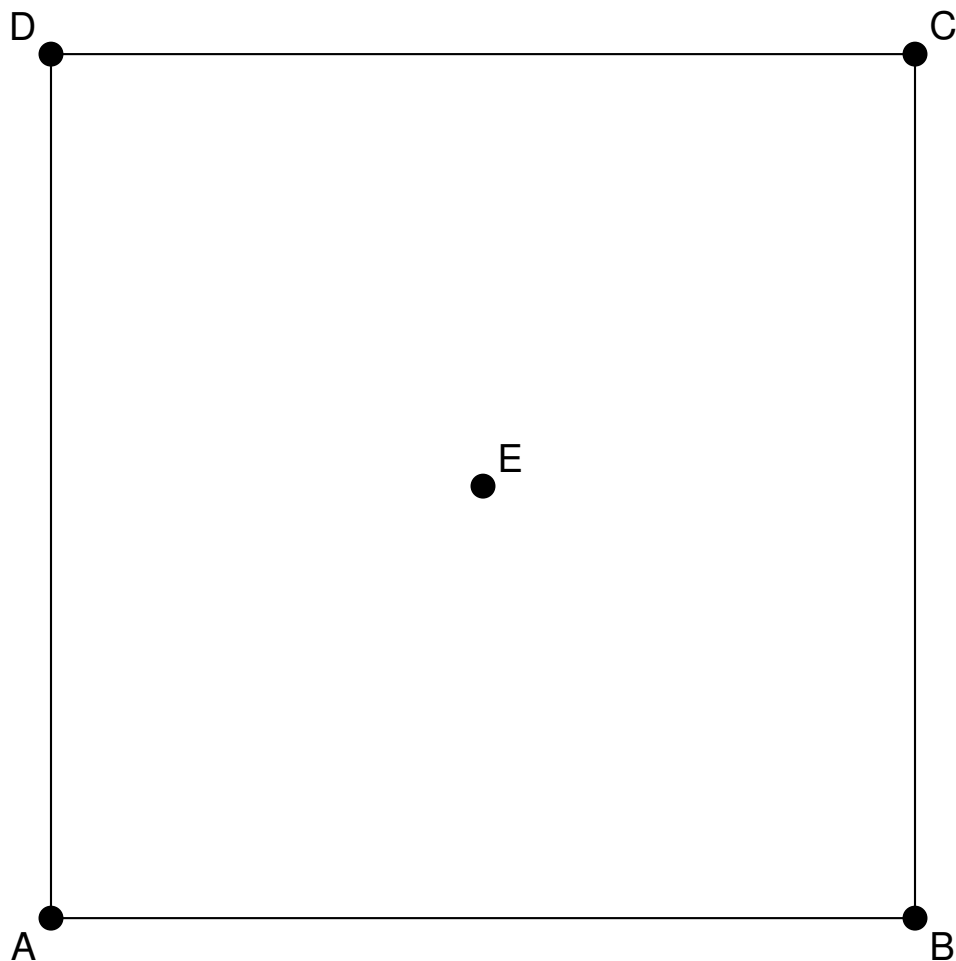
For Problems 3 and 4, use straight-line distances. New points must stay inside or on the square. If nearest dots tie, their shared distance counts.

Problem 3: Put a new point in the square so its distance to the nearest black dot is as large as possible. Find every best place, and explain why no other place is better.



Place	Distance to its nearest black dot

Problem 4: The center is now a black dot too. Find every best place in the square, and explain why no other place has a larger distance to its nearest black dot.



Place	Distance to its nearest black dot